



MAGIC Gamma Telescope Prediction Using Machine Learning

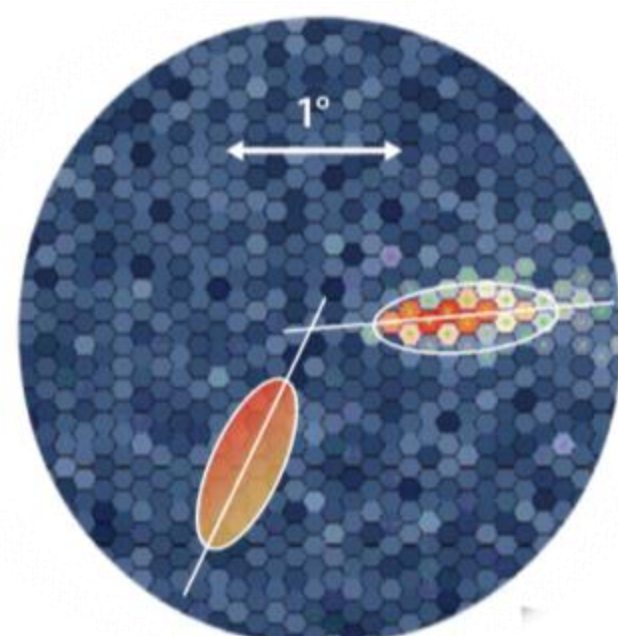
Simulation of High Energy Gamma Particles in Cherenkov Telescope

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Introduction

The Cherenkov gamma telescope observes high energy gamma rays by capturing Cherenkov light, radiation produced when gamma rays interact with the Earth's atmosphere, in the detector. The light is then recorded as an image on a camera, which contains patterns consisting of hundreds to thousands of collected Cherenkov photons and resembles elliptical projections. These images can be analyzed to distinguish core gamma-ray events from images of hadronic showers caused by other rays in the upper atmosphere.



Purpose

In this study, data was generated using a Monte Carlo simulation, which produces random samples based on a physical distribution. This data was then analyzed using key image variables, such as length, concentration, and asymmetry, and used to train machine learning models with the goal of classifying these events accurately and improving gamma-ray detection.

Methods

The study was carried out by deploying nine different machine learning models through a data split of 70% training and 30% testing. The most optimal model was selected from among the following:

- Decision Tree, Tuned Decision Tree
- Bagging Classifier, Tuned Bagging Classifier
- Random Forest, Tuned Random Forest
- Logistic Regression
- XGBoost, Tuned XGBoost

Since both false negatives and false positives were disruptive to the study, the model was set to maximize the **F1 score** to balance precision and recall.

Dataset

The MAGIC Gamma Telescope data was taken from the UC Irvine Machine Learning Repository, which consists of 19020 rows and 11 columns. Although the data was skewed to the right, this was not addressed as it was not considered to significantly affect the results in the end.

Independent Variables:

- Figure Length: Major Axis of Ellipse (mm) - I/R
- Figure Width: Minor Axis of Ellipse (mm) - I/R
- Figure Size: 10-log of Total Light Collected (phot) - I/R
- Figure Concentration: Ratio of Sum of Two Highest Pixels Over Total Pixels – I/R
- Figure Concentration 1: Ratio of Highest Pixel Over Total Pixels – I/R
- Figure Asymmetry: Distance from Highest Pixel to Center, Projected onto Major Axis (mm) - I/R
- Figure Moment 3 Longitudinal: Skewness of Ellipse Over Length (mm) - I/R
- Figure Moment 3 Transverse: Skewness of Ellipse Over Width (mm) - I/R
- Figure Alpha: Angle between Ellipse and Camera (deg) - I/R
- Figure Distance: Distance of Ellipse Center and Camera Center (mm) - I/R

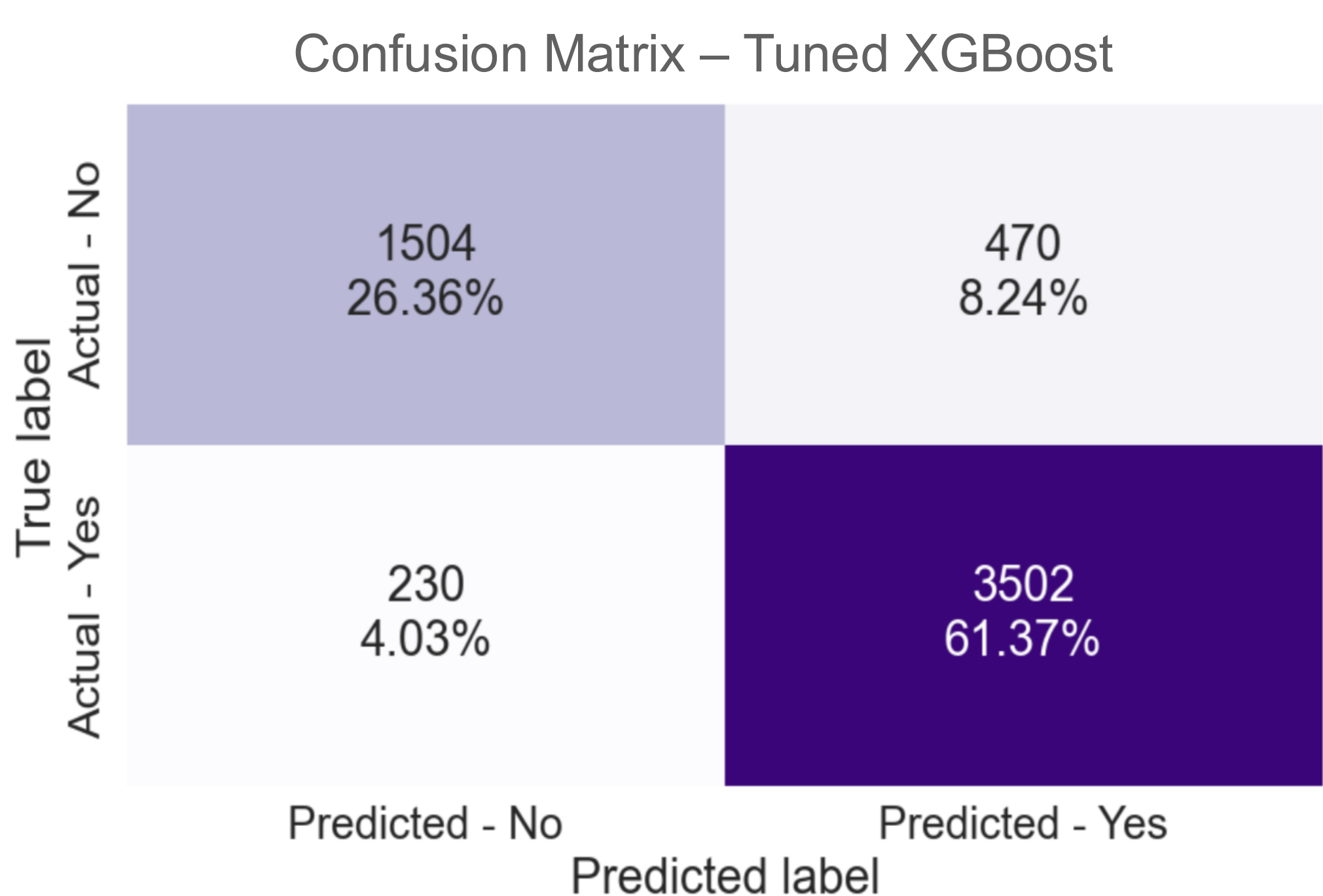
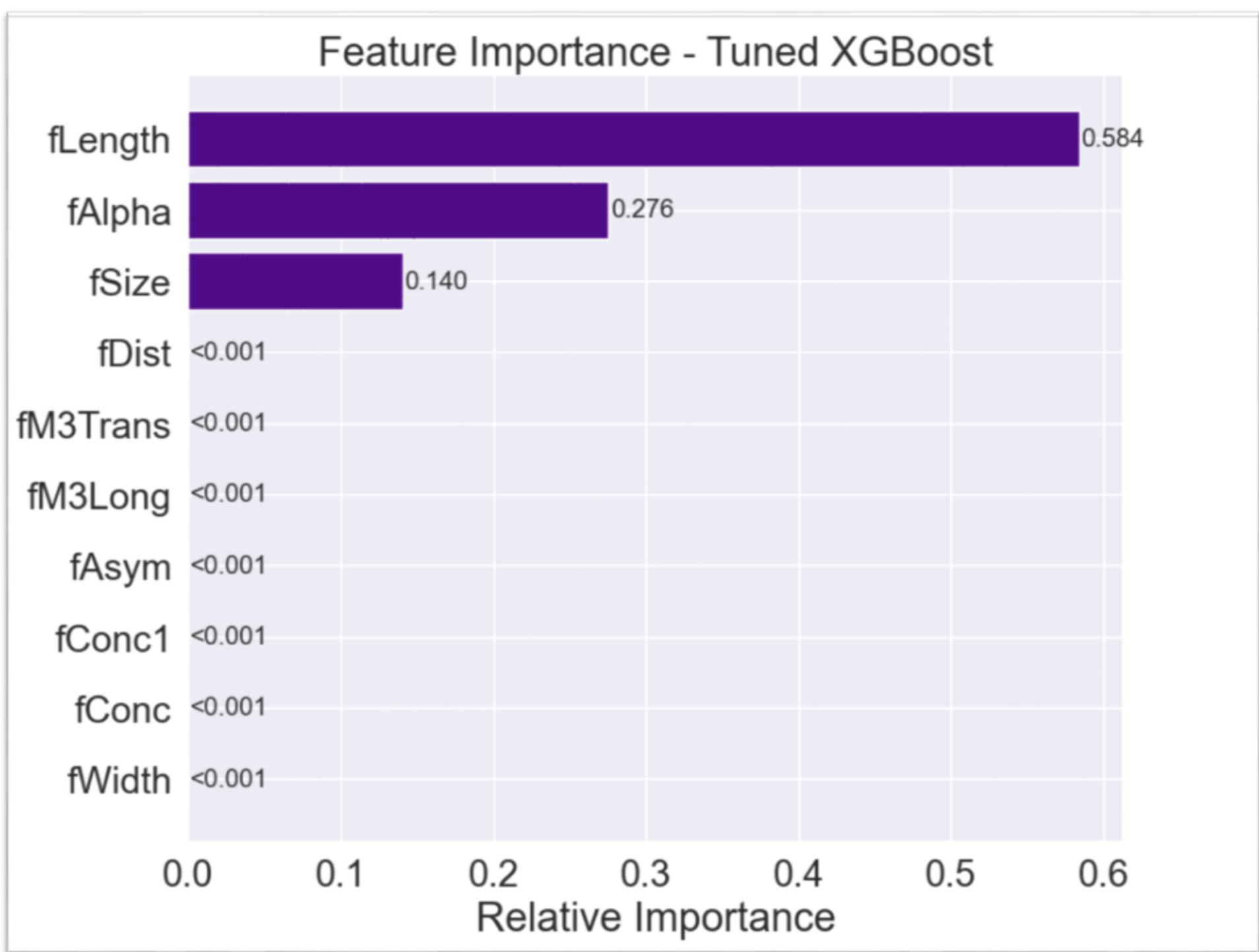
Dependent Variables:

- Outcome (gamma or hadronic) - Nominal
 - Variable was converted to an integer column: gamma – 1, hadronic - 0

Results

The Tuned XGBoost model was chosen because it had the highest F1 test score and performed well on other metrics as well.

Model	Train_Accuracy	Test_Accuracy	Train_Recall	Test_Recall	Train_Precision	Test_Precision	Train_F1	Test_F1
Tuned XGBoost	0.903	0.879	0.964	0.947	0.894	0.878	0.928	0.911
XGBoost	0.974	0.877	0.995	0.938	0.965	0.882	0.980	0.909
Random Forest	1.000	0.875	1.000	0.936	1.000	0.880	1.000	0.907
Bagging Classifier	0.994	0.862	0.996	0.906	0.995	0.886	0.995	0.896
Decision Tree	1.000	0.816	1.000	0.861	1.000	0.858	1.000	0.859
Logistic Regression	0.793	0.787	0.899	0.899	0.804	0.800	0.849	0.847
Tuned Decision Tree	0.646	0.654	1.000	1.000	0.646	0.654	0.785	0.791
Tuned Bagging Classifier	0.646	0.654	1.000	1.000	0.646	0.654	0.785	0.791
Tuned Random Forest	0.646	0.654	1.000	1.000	0.646	0.654	0.785	0.791



Discussion & Insights

- The Tuned XGBoost model performed the best in the Test_F1 category and performed well in every other metric
- The length of the major axis of the ellipse was the most defining feature of gamma-ray events
- Despite testing different methods to address skewness with guidance from ChatGPT, no significant differences were found in the end results. However, it is possible that certain transformation techniques could further improve the model's predictive performance.
- The dataset was Monte Carlo generated, meaning a simulation using real-world data could potentially differ.

Summary

- **Selected Model:**
 - Tuned XGBoost
- **Most Important Evaluation Metric:**
 - F1 Score
- **Most Relevant Features:**
 - Figure Length, Figure Alpha, and Figure Size

References

Bock, R. (2004). MAGIC Gamma Telescope [Dataset]. UCI Machine Learning Repository. <https://doi.org/10.24432/C52C8B>.

Cherenkov Telescope Array at uzh. Physik. (n.d.). <https://www.physik.uzh.ch/en/researcharea/cta.html>

